Quality Management

9 — Camera sensors: from photons to grey levels

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Today

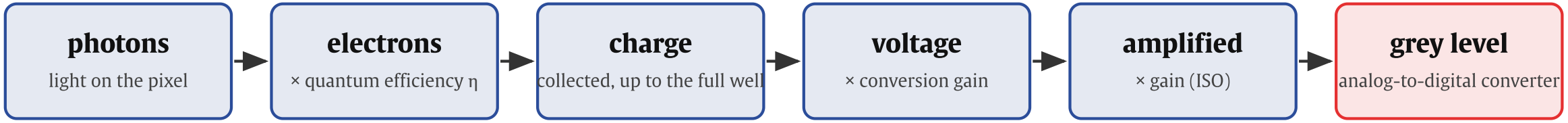
The link between the lens and the numbers

Lectures 7 and 8: the lens forms an image, and the image becomes a matrix of numbers. Today: the **sensor** that does the conversion — how it works, and what limits every measurement made with it.

1. **Photons to electrons**
   photodiode, pixel, quantum efficiency
2. **Pixels to an image**
   CCD and CMOS, shutters, line and area
3. **Colour, spectrum, size**
   Bayer filter, IR, sensor formats
4. **What limits it**
   noise, saturation, dynamic range
5. **Choosing a camera**
   for the washers, and for a line

The sensor is a measurement chain

Five conversions between the light and the grey level



- **Quantum efficiency η** — electrons generated per photon arriving, between 0 and 1. Typical: 0.5–0.8 in green light.
- **Full-well capacity** — the most electrons a pixel can hold before it saturates. About 10 000 for a 3.45 µm pixel.
- **Conversion gain** — volts per electron, then grey levels (DN, digital numbers) per electron after the converter.

One standard to compare cameras

EMVA 1288 defines how machine-vision cameras are measured and what their datasheets must state: quantum efficiency, read noise, full well, dark current, non-uniformity. Two cameras compared on EMVA data are compared fairly.

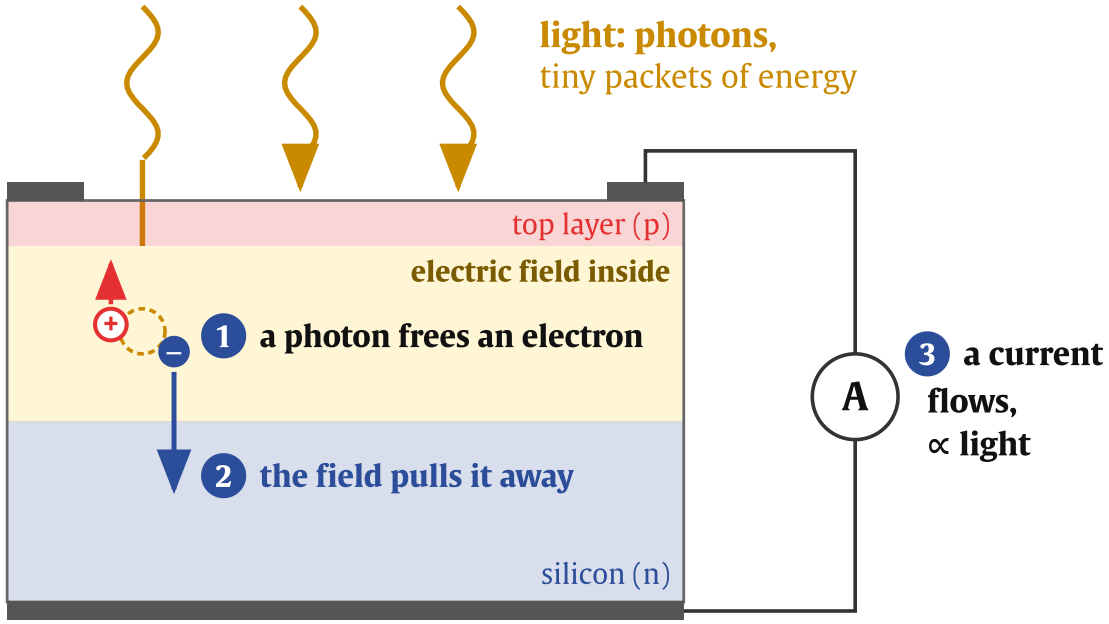
PART 1

From photons to an image

Photodiode · Pixel · CCD and CMOS · Shutters · Line and area · Colour · Spectrum · Size

The photodiode

Every pixel is one



a pixel is one photodiode: it stores the electrons instead of measuring the current

- **What it is** — a small piece of silicon that turns light into an electric current. Every pixel of a camera sensor is one.
- **How it works** — light is made of tiny packets of energy, **photons**. A photon entering the silicon knocks an **electron** free; an electric field inside the diode pulls it away, and it flows out as a current. Silicon sees up to about 1.1 µm.
- **Why it matters** — the current is **proportional to the light**: twice the light, twice the current. So it measures light — in light meters, barcode readers, and every camera.

$$\text{electrons per second} = \eta \times \text{photons per second}$$

η, the **quantum efficiency**: the share of photons that free an electron, 0.5–0.8.

HOW MANY ELECTRONS

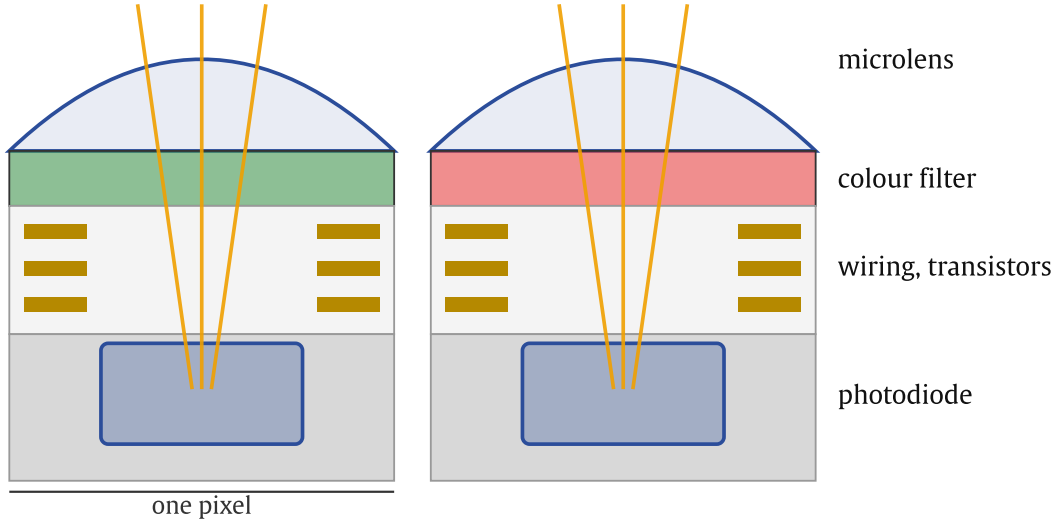
1 nW of green light — a very faint light — is 2.8 billion photons per second. With $\eta = 0.6$: 1.7 billion electrons per second, a current of **0.27 nA**.

In a camera: a bucket, not a meter

A pixel does not measure the current: like a **bucket collecting rain**, it stores the electrons during the exposure, and they are counted at the end. The bucket's size is the full well.

Inside a pixel

A photodiode, a filter and a lens, a few micrometres wide



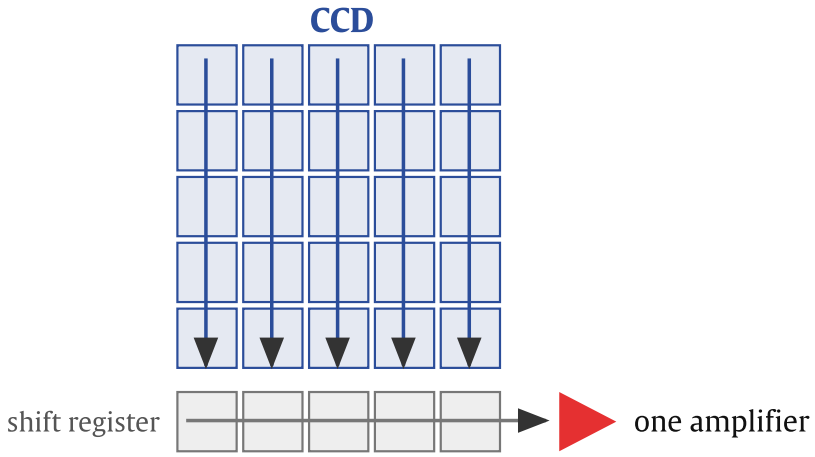
- **Microlens:** focuses the light onto the photodiode, so light falling on the wiring is not lost.
- **Colour filter:** on colour sensors only — one colour per pixel (slide 12).
- **Photodiode:** collects the electrons during the exposure. The fraction of the pixel's area that is sensitive is the **fill factor**.
- **Wiring and transistors:** read the charge out. In *back-side illuminated* sensors they sit below the photodiode, so they no longer shade it.

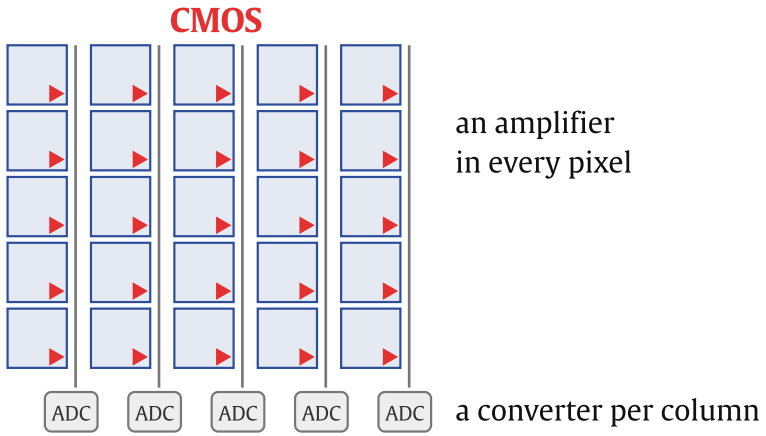
Pixel size

Phones: 0.6–1.4 µm. Machine vision: 2–6 µm. Larger pixels collect more light and hold more electrons: less noise (Part 2).

Reading the pixels: CCD and CMOS

The difference is where the charge becomes a voltage





CCD — charge-coupled device

The charges are shifted, row by row, to **one** output amplifier. Every pixel is read by the same electronics: very uniform and low-noise, but slow and power-hungry. Production has largely ended.

CMOS — complementary metal-oxide semiconductor

Every pixel has its **own amplifier**, and each column its own converter. Fast, low power, cheap — and it can read just a **region of interest**: half the rows, about twice the frame rate. Today almost every industrial camera is CMOS.

CCD or CMOS?

Strengths, weaknesses, and where each is used

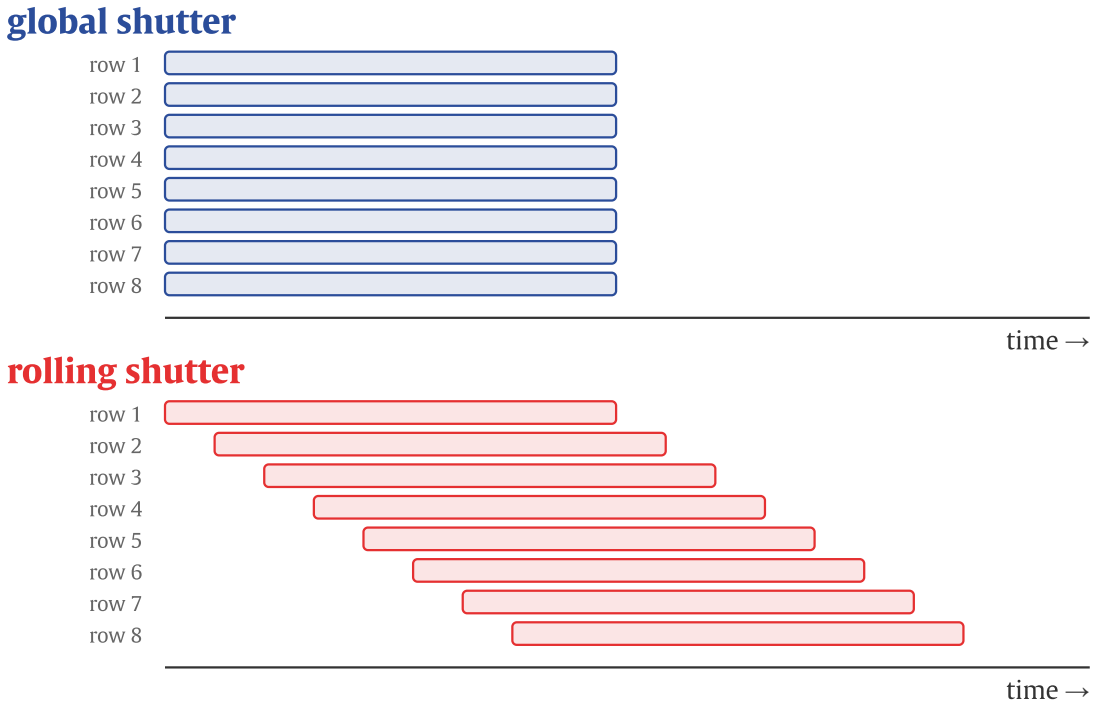
	CCD — charge-coupled device	CMOS — active-pixel sensor
Charge to voltage	at one amplifier, at the corner of the chip	in every pixel
Speed	slow: every pixel passes through one output	fast: columns read in parallel — hundreds to thousands of images per second
Region of interest	no faster: all the charge must still be shifted out	read only the rows you need, and go faster
Uniformity	very uniform: one amplifier for every pixel	small differences from pixel to pixel, removed by calibration (slide 22)
Read noise	low: a few electrons, at slow readout	today as low or lower: 1–3 electrons
Shutter	global (interline-transfer CCDs)	mostly rolling; global in many industrial models
Bright spots	blooming and smear: bright vertical streaks (slide 23)	little blooming, no smear
Power and cost	high: several supply voltages, timing and converter on separate chips	low: everything on one chip, made like any logic chip
Where it is used	scientific cameras with long exposures — astronomy, spectroscopy; older inspection systems still in service	phones, cars, machine vision, high-speed cameras — almost every camera made today

Why CMOS won

For decades the CCD gave the cleaner image. Around 2010 CMOS caught up in noise and uniformity, while staying faster, cheaper and less power-hungry. Most manufacturers have since stopped making CCDs.

Global and rolling shutter

Do all rows look at the scene at the same moment?



- **Global shutter:** all rows start and stop collecting light together. A moving part is recorded as it was at one instant.
- **Rolling shutter:** each row starts a little after the one above. Cheaper and common in CMOS — and in every phone. A moving part is **skewed**.

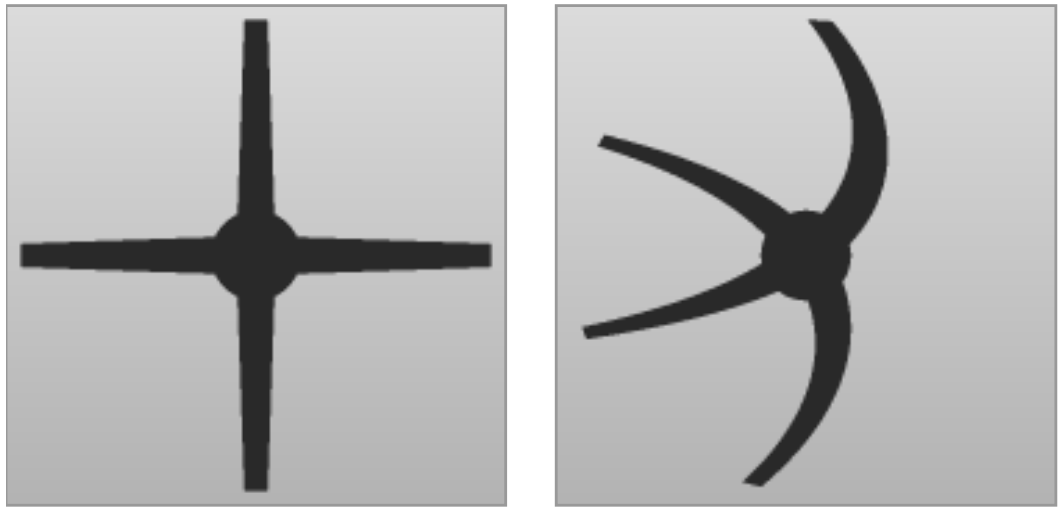
For moving parts

A global shutter, or a rolling shutter with a **flash** fired while all rows collect. For still washers on a light box, a phone's rolling shutter is fine.

Rolling shutter

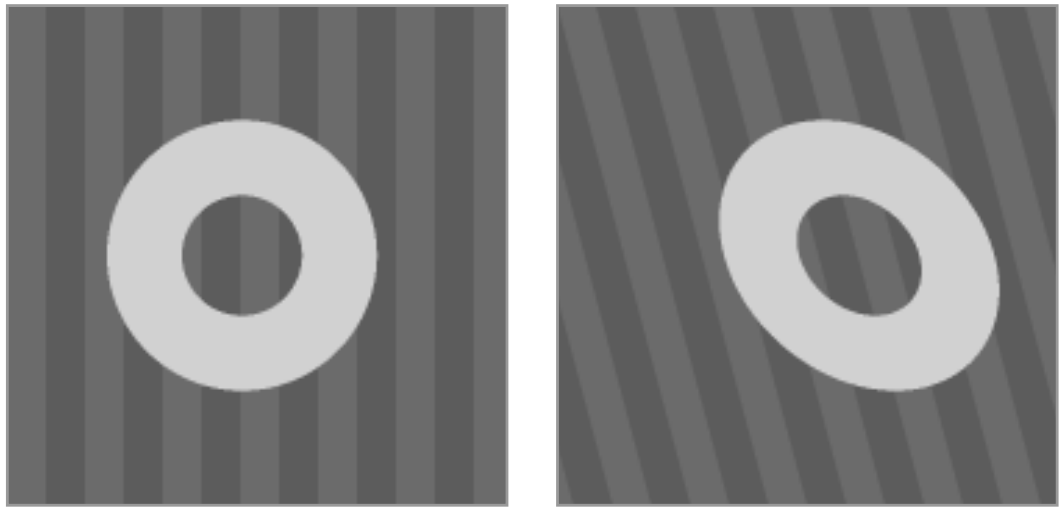
The same moving scenes, recorded both ways

a rotor turning: 0.3 turn while the rows are read



global shutter | rolling shutter

a washer on a conveyor, moving right



global shutter | rolling shutter

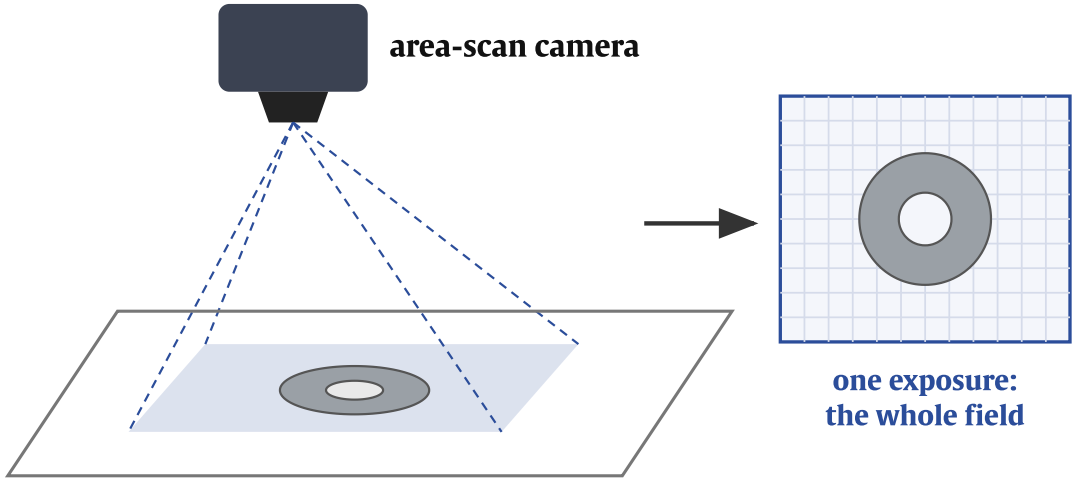
- **Rotor:** each row sees the blades a little later, after they have turned further. Straight blades come out bent, and can look detached from the hub — the familiar phone photo of a helicopter or a propeller.
- **Conveyor:** the lower rows see the washer further along. The round washer comes out slanted, and the belt's straight slats lean.
- **For a measurement:** a gauge would find the washer out of round — a systematic error that grows with the speed of the line.

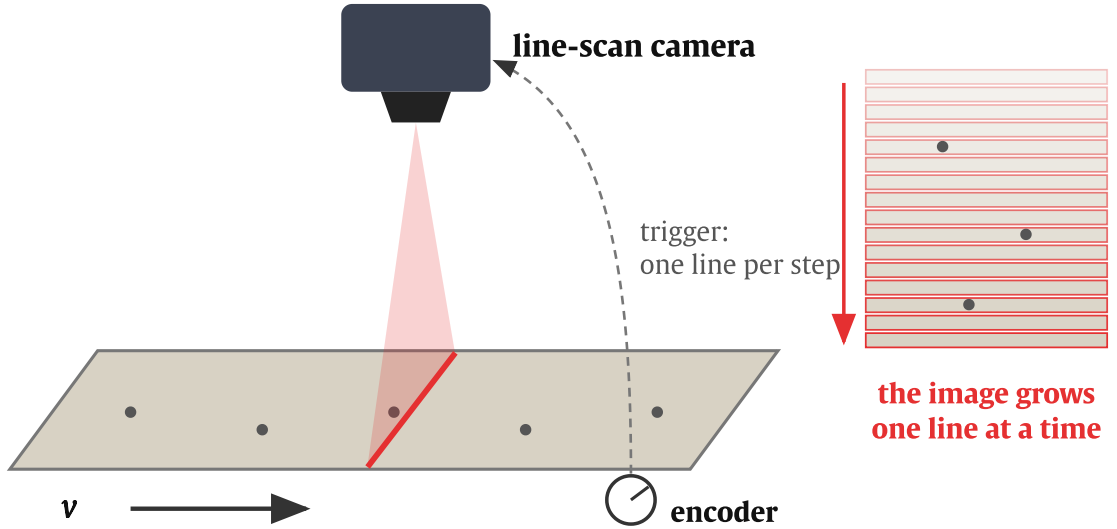
ON A CONVEYOR

Reading 2048 rows takes 20 ms. A part moving at 0.5 m/s travels **10 mm** between the first row and the last: a round washer comes out as an ellipse.

Area-scan and line-scan cameras

A grid of pixels, or one line and a moving part





Area-scan

A two-dimensional grid: the whole field in one exposure. For discrete parts — washers, castings, assemblies.

Line-scan

One row of pixels. The image is built line by line **as the part moves past**, each line triggered by an encoder. For continuous material — paper, sheet steel, textiles, glass — and cylinders turned in front of the camera.

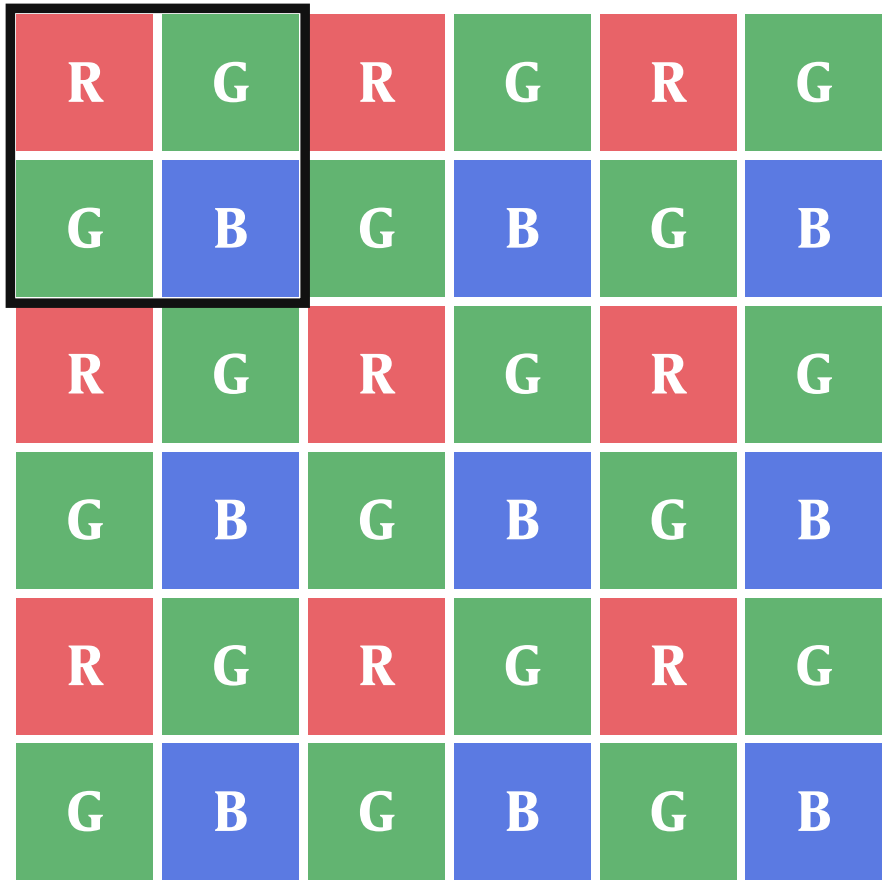
INSPECTING A STEEL STRIP

1 m wide, one line of 8192 pixels: **0.12 mm** per pixel across. At 2 m/s, square pixels need a line every 0.12 mm: **16 400 lines per second**.

The encoder ties each line to a distance, not to a time: if the line slows down, the image does not stretch. **TDI** (time-delay integration) cameras add up several rows as the part moves past: more light per line on fast lines.

Colour sensors

A photodiode does not see colour: a filter in front of it does



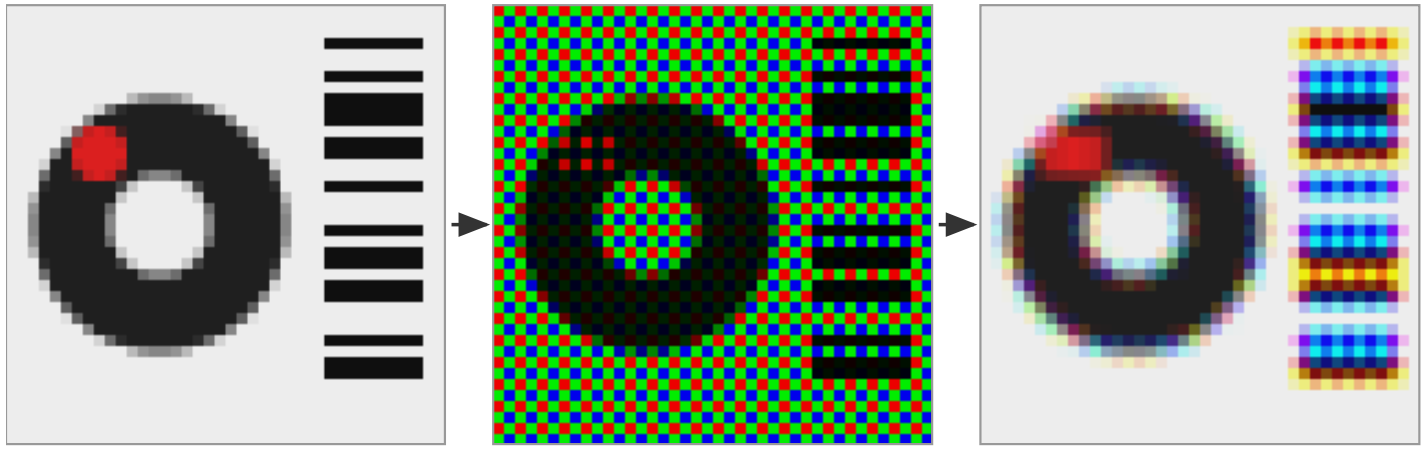
half the pixels green, a quarter red, a quarter blue

- **Bayer filter:** a mosaic of red, green and blue filters, one per pixel, in the pattern R G / G B. Green appears twice because the eye is most sensitive to it — it carries most of the brightness.
- **Demosaicing:** each pixel records one colour; the other two are **estimated from its neighbours** (next slide).
- **Three-chip cameras:** the light is split onto three sensors, one per colour (next slide).

> **For measurement: monochrome**
> No filters: every pixel records the full signal, with two to three times more light. Gauging cameras are monochrome; your phone is a Bayer camera (lab exercise 1).

Three colours per pixel: two ways

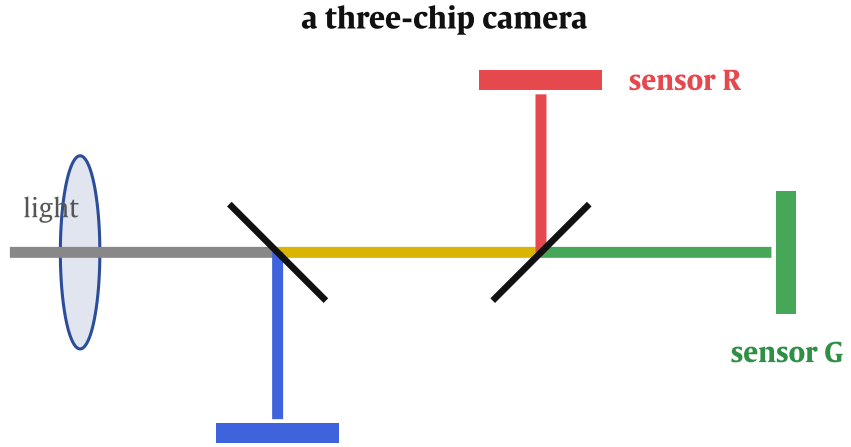
Estimate two of them from the neighbours, or split the light onto three sensors



1 · the scene

2 · what the sensor records
one colour per pixel

3 · after demosaicing
false colours on every edge



dichroic mirrors: each reflects one colour, passes the rest

- **Demosaicing** (1 → 3): each pixel keeps its own colour and takes the other two from its neighbours. Where the image changes slowly the estimate is good; at an edge the neighbours disagree, and **false colours** appear — around the grey washer and on the barcode's thin bars.
- **For measurement:** the false colours sit exactly on the edges a gauge measures, and each colour is sampled more coarsely than the pixel count suggests — red and blue on one pixel in four.

Three-chip camera
Mirrors split the light by colour onto three sensors: all three colours at every pixel, no fringes. Used where colour must be exact — print and food inspection, broadcast video — but three sensors cost more and must stay aligned to a fraction of a pixel.

What the sensor can see

Beyond the visible — and why that is useful on a line

Range	Sensor	Use in quality control
Visible, 400–700 nm	silicon, with an **IR-cut filter**	what people see: colour, marking, assembly
Near infrared, 700–1000 nm	silicon, **without** the IR-cut filter	lighting invisible to operators, and insensitive to the room's light
Short-wave IR, 0.9–1.7 µm	InGaAs	moisture in food and wood, sorting plastics, seeing through silicon wafers
Thermal IR, 8–14 µm	microbolometer	heat, not light: welds, electrical faults, insulation, hidden defects
Ultraviolet	UV-enhanced silicon	fluorescent markings, cracks shown by penetrant inspection, coatings

The IR-cut filter

Silicon sees up to 1.1 µm, the eye only to 0.7. Without the filter, a colour camera mixes infrared into its colours and the image looks wrong: consumer cameras always have one.

Choose the light first

The sensor's range only matters together with the light: a near-infrared LED and a filter on the lens make an inspection that ignores the sunlight through the factory windows.

Sensor size and format

The names are historical; the dimensions matter

Format	Diagonal	Typical use	Crop factor
1/2″	8 mm	machine vision, phones	5.4
2/3″	11 mm	machine vision	3.9
1″	16 mm	machine vision, compacts	2.7
APS-C	about 28 mm	cameras	about 1.5
Full frame	43.3 mm (36 × 24)	cameras	1

"1/2 inch" is not a dimension of the sensor: it comes from the diameter of the old television tubes that sensors replaced. Read the size in millimetres. **Crop factor** = 43.3 mm / diagonal: the same lens on a smaller sensor sees a narrower field.

THE WASHER CAMERA

2448 × 2048 pixels of 3.45 µm: 8.45 × 7.07 mm, diagonal 11.0 mm — a **2/3″** sensor.

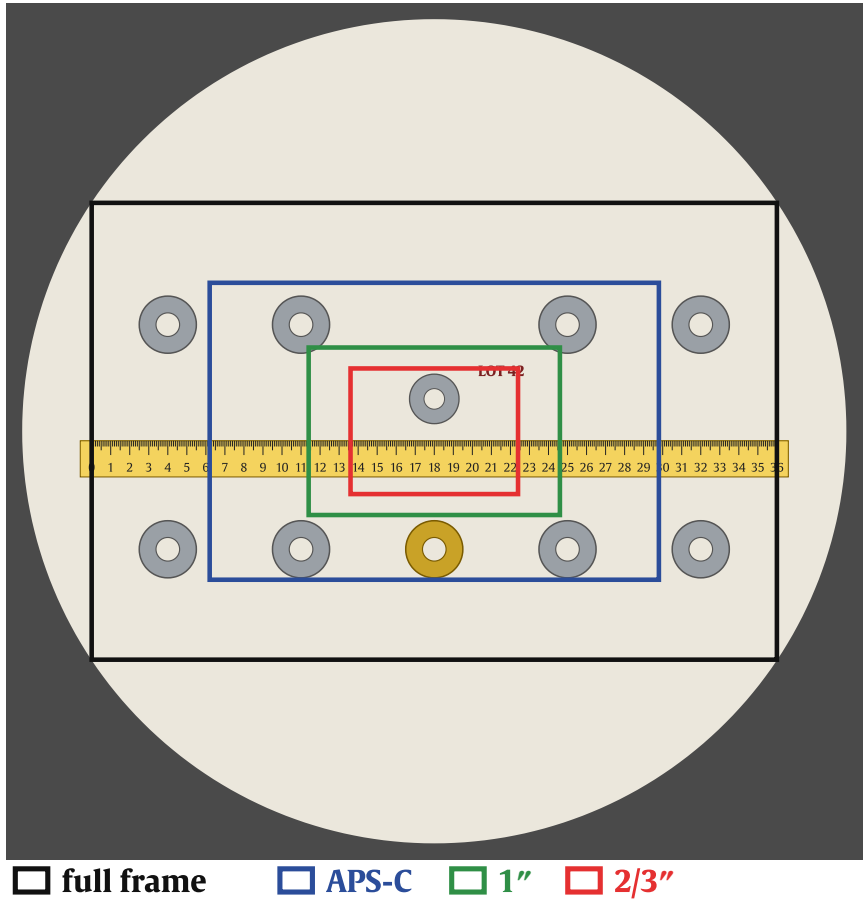
The lens must cover the sensor

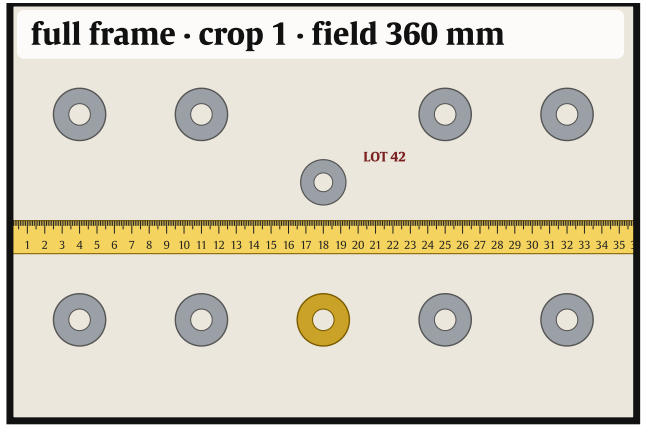
Every lens projects a circle of light. If it is smaller than the sensor's diagonal, the corners go dark. Machine-vision lenses are sold by format: a 2/3″ lens for a 2/3″ sensor.

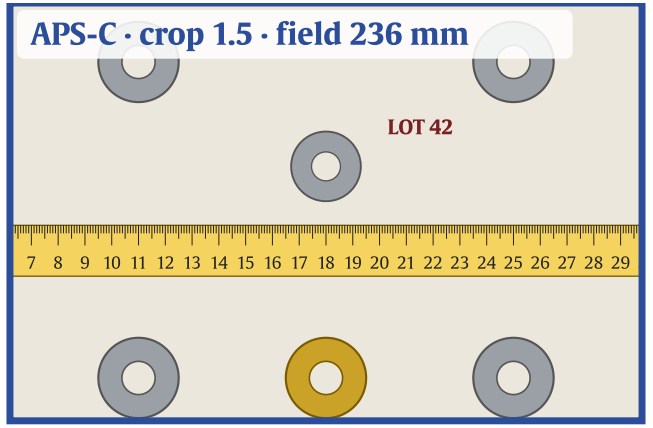
More pixels on the same area means smaller pixels: less light each, fewer electrons, more noise.

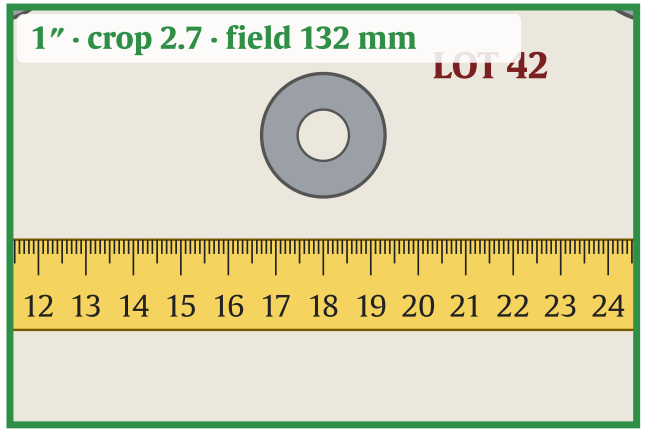
Crop factor

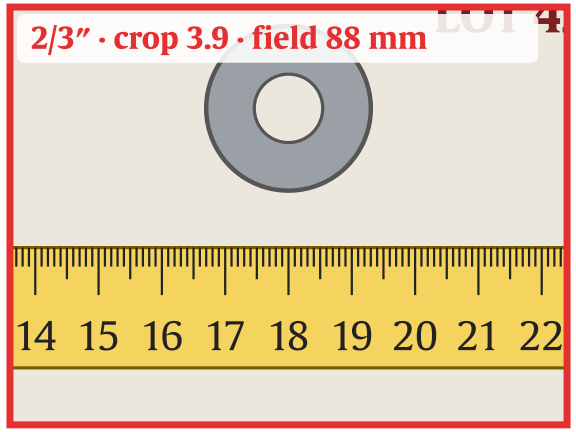
One lens, four sensors: each records only its own part of the image











The lens forms the same image whatever sits behind it. A smaller sensor cuts out a smaller piece; shown at the same size, it looks **magnified**: fewer washers, larger. **Why it matters:** the field of view depends on lens and sensor together — choose the lens for your sensor.

EQUIVALENT FOCAL LENGTH

Focal length × crop factor: a 25 mm lens on the 2/3″ washer camera frames like a 98 mm lens on full frame.

Recap TAKEAWAYS

Part 1 — from photons to an image

1. **The chain:** photons → electrons (quantum efficiency) → charge (up to the full well) → voltage → grey levels. EMVA 1288 is how cameras are compared.
2. **Photodiode:** current proportional to light, up to about 1.1 µm in silicon. A pixel collects and counts electrons during the exposure.
3. **CCD vs CMOS:** one amplifier for all, or one per pixel. CMOS is fast, reads regions of interest, and is what industry uses today.
4. **Moving parts:** global shutter or a flash; continuous material: a line-scan camera triggered by an encoder.
5. **Colour, spectrum, size:** Bayer cameras estimate two colours out of three; measure in monochrome. Choose the spectral range with the light. Read sensor sizes in millimetres.

PART 2

What limits the sensor

Noise · Signal-to-noise ratio · Calibration · Saturation · Dynamic range · Linearity

Where the noise comes from

Every source in the same unit: electrons, for one camera and one exposure

The washer camera (full well 10 000 e^-, read noise 5 e^-, 12 bits), a 10 ms exposure at 25 °C. A **bright** pixel holds 5000 electrons, a **dark** one 50. Every size is a standard deviation, in electrons.

Source	Cause	How it grows	Bright pixel 5000 e^-	Dark pixel 50 e^-	What reduces it
Shot noise	light arrives as photons, at random	$\sqrt{N}$	**71**	**7**	more light — nothing else
Dark current	electrons freed by heat: about 10 per second here	with exposure time; doubles every 6–8 °C	0.3	0.3	short exposures, cooling
Read noise	the amplifier and the converter	the same at every read	5	**5**	a better camera
Quantisation	rounding to grey levels: one level is 2.4 e^-	$\Delta / \sqrt{12}$	0.7	0.7	more bits — with 8 bits it is 11
Fixed pattern	each pixel's own offset and gain	about 1 % of the signal, plus 2 e^-	50	2	**calibration** (slide 22)
Random total	the first four, combined as √(sum of squares)		**71** · SNR 70	**9** · SNR 6	

Bright pixels: shot noise is the whole story — only more light helps. **Dark pixels:** read noise is as large as shot noise — there, a better camera helps.

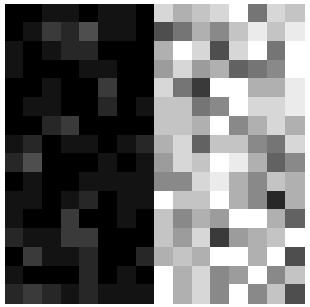
The first four are **random**: they change from image to image and average out. Fixed pattern is **systematic**: as large as shot noise in bright pixels, but the same in every image — measured once, it is removed.

Signal-to-noise ratio

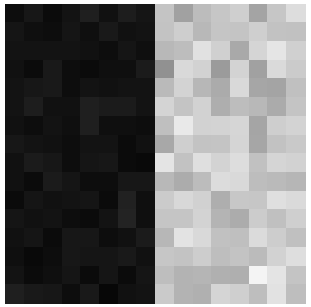
Shot noise sets the limit: four times the light for twice the quality

$$\mathrm{SNR} = N / \sqrt{(N + \sigma_{\mathrm{read}}{}^2 + \sigma_{\mathrm{dark}}{}^2)}$$

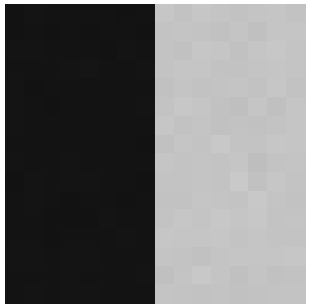
In good light shot noise dominates and **SNR ≈ √*N***. A pixel holding 10 000 electrons can reach SNR = 100 at most; doubling the SNR takes four times the light.



10 electrons
SNR ≈ 3



100 electrons
SNR ≈ 10



10 000 electrons
SNR ≈ 100

THE WASHERS, READ NOISE 5 ELECTRONS

Background at 80 % of the full well: 8000 electrons, SNR ≈ 89. Washer at 8 %: 800 electrons, SNR ≈ 28. Both far above what a threshold needs.

Averaging

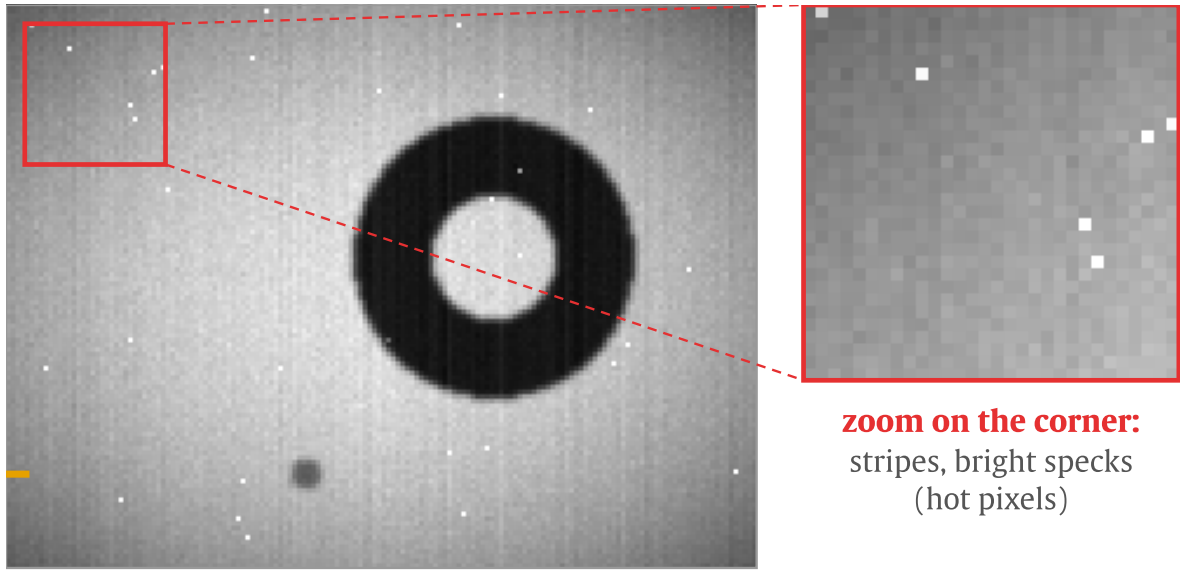
Averaging k independent values divides random noise by $\sqrt{k}$ — over pixels, or over images. It is why fitting hundreds of edge points beats a pixel (lecture 12), and why you take several photos in the lab.

Gain does not help

Raising the gain (ISO) multiplies signal and noise together: the SNR stays the same, and the brightest values saturate sooner. More light is the only real fix.

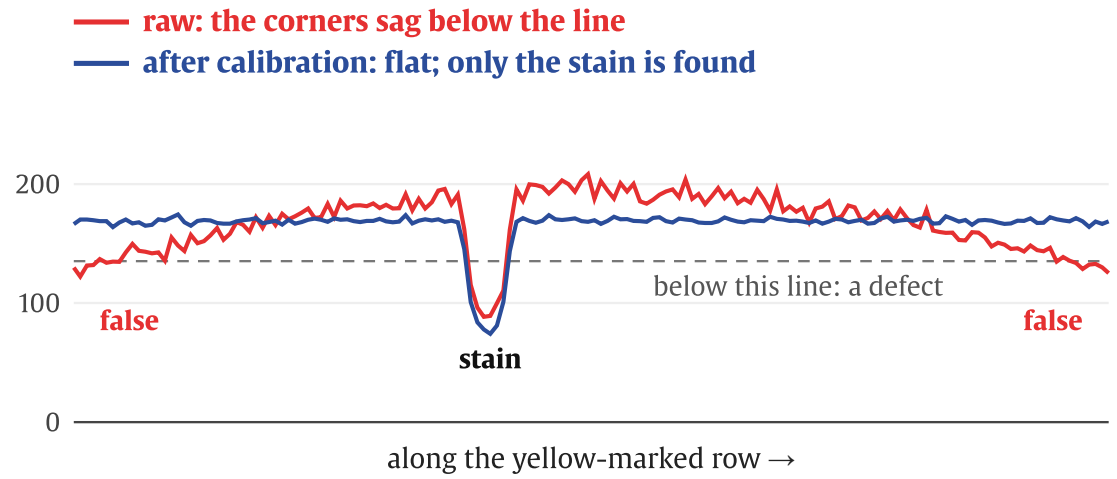
Fixed-pattern noise

Every pixel has its own small error — the same in every image



raw image: the corners are darker

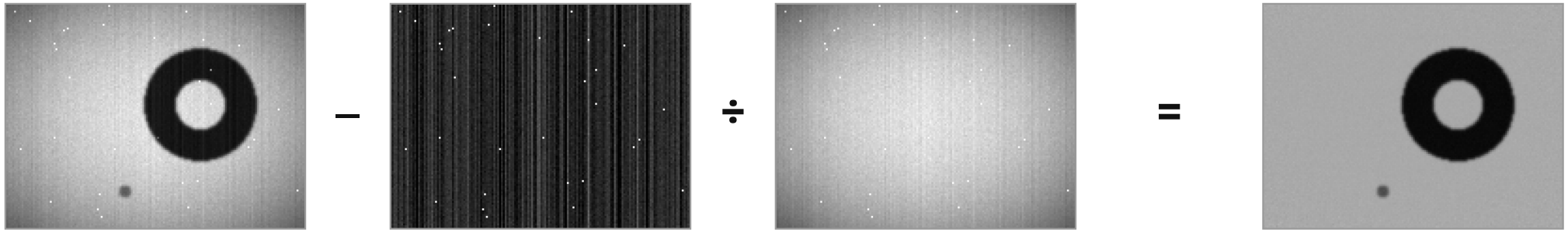
zoom on the corner:
stripes, bright specks
(hot pixels)



- **What it is** — each pixel has its own small **zero error** (offset) and its own **sensitivity** (gain); a few **hot pixels** read bright even in the dark; and the lens makes the corners darker (vignetting). Unlike random noise, the pattern is the same in every image.
- **Why it matters** — a fixed threshold finds **false defects** in the darker corners and puts edges in the wrong place. Averaging many images does not help: it averages the same pattern.
- **How to remove it** — because it repeats, it can be measured once and taken out of every image: a dark frame and a flat field (next slide).

Calibrating the camera: dark frame and flat field

Measure each pixel's zero and span once, then correct every image



raw image I
the washer, as recorded

dark frame D
lens covered · shown ×6

flat field F
the empty light box

corrected I_c
uniform: ready to measure

Dark frame *D* — each pixel's zero
Images taken with **no light at all**: lens cap on, same exposure time and temperature as the measurement. What is left is each pixel's offset, and the hot pixels. Average 10–20 of them, so random noise does not enter the calibration.

Flat field *F* — each pixel's span
Images of a target that is **equally bright everywhere** and fills the whole field: for the washers, the **empty light box**; otherwise a white diffuser plate. Same lens, aperture, focus and exposure, at about two thirds of full scale. The target is uniform, so every difference in the image is the camera's: darker corners and each pixel's gain. Average 10–20.

$$I_c = (I - D) \,/\, (F - D) \cdot \mathrm{mean}(F - D)$$

Subtract the zero, divide by the span, pixel by pixel — like calibrating any instrument. Redo it when temperature, exposure time, lens or aperture change.

Saturation and blooming

When a pixel is full, the information is gone — and can spill over



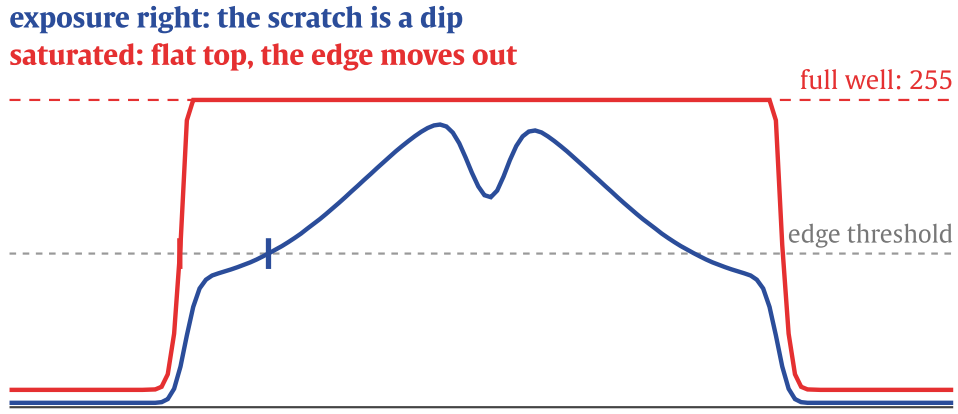
exposure right
the scratch crosses the reflection



4 × longer: saturated
the scratch is gone



4 × longer, on a CCD
blooming: charge runs up and down



grey level along the yellow-marked row

- **Saturation:** a pixel whose bucket is full — its full well — stops counting. Everything brighter reads the same: the scratch in the washer's reflection is **lost**, not just compressed — and the edge next to it moves outwards.
- **Blooming:** in a CCD the extra charge spills into the neighbouring pixels along the column — a bright streak. Anti-blooming drains and CMOS sensors largely prevent it.

What to do

Diffuse light (a dome or a light tent), a **polarising** filter on the light and on the lens, or a shorter exposure. Check the histogram of every new setup: no peak at the top value.

Dynamic range

The brightest and the darkest thing one image can hold

$$\text{DR} = \text{full well} \,/\, \text{read noise}$$

$$\text{DR}_{\text{dB}} = 20 \log_{10}(\text{full well} \,/\, \text{read noise})$$

The ratio between the largest signal and the smallest one above the noise floor. In bits: $\log_2(\text{DR})$ — the number of bits worth recording.

THE WASHER CAMERA

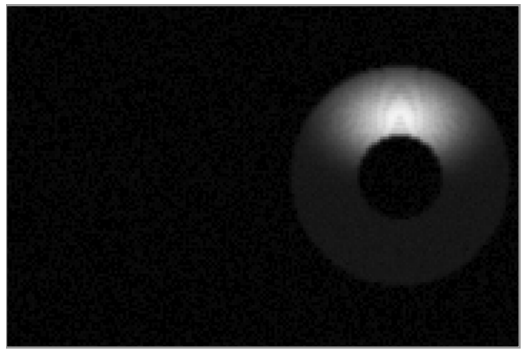
10 000 electrons / 5 electrons = 2000: **66 dB**, about **11 bits**. A 12-bit converter records all of it; 8 bits throw some of it away — which is fine when, as on the light box, the scene needs far less.

When the scene exceeds it

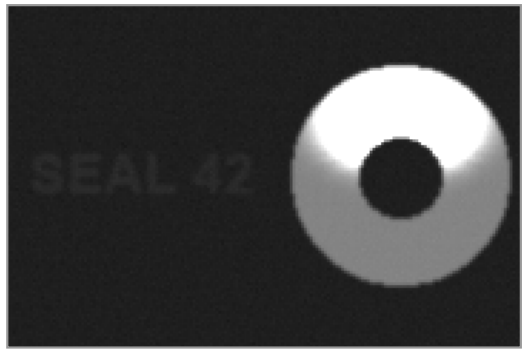
A shiny washer on a black rubber seal can need more range than the sensor has (next slide). Then: change the lighting first; otherwise merge several exposures into one (HDR, next slide).

When the scene has more range than the sensor

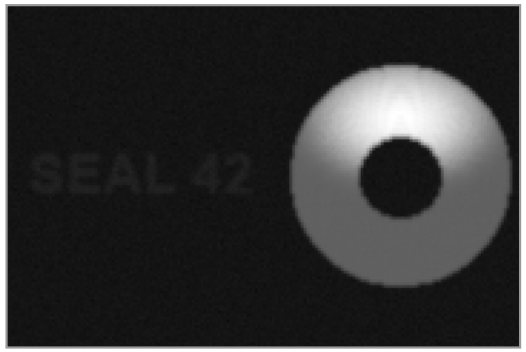
A shiny washer on a black rubber seal



short exposure
washer fine · text lost in the noise

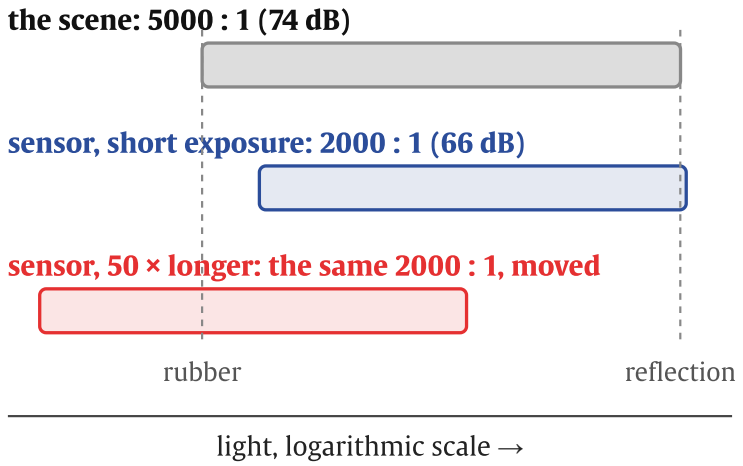


50 × longer
text readable · washer saturated



the two merged (HDR)
both: the range of two images

all three shown with the same tone curve



Definition · HDR (high dynamic range) imaging

Several images of the same scene at **different exposure times**, merged pixel by pixel: each pixel is taken from the image in which it is neither saturated nor lost in the noise, then divided by that image's exposure time — so every pixel ends up on one scale of light.

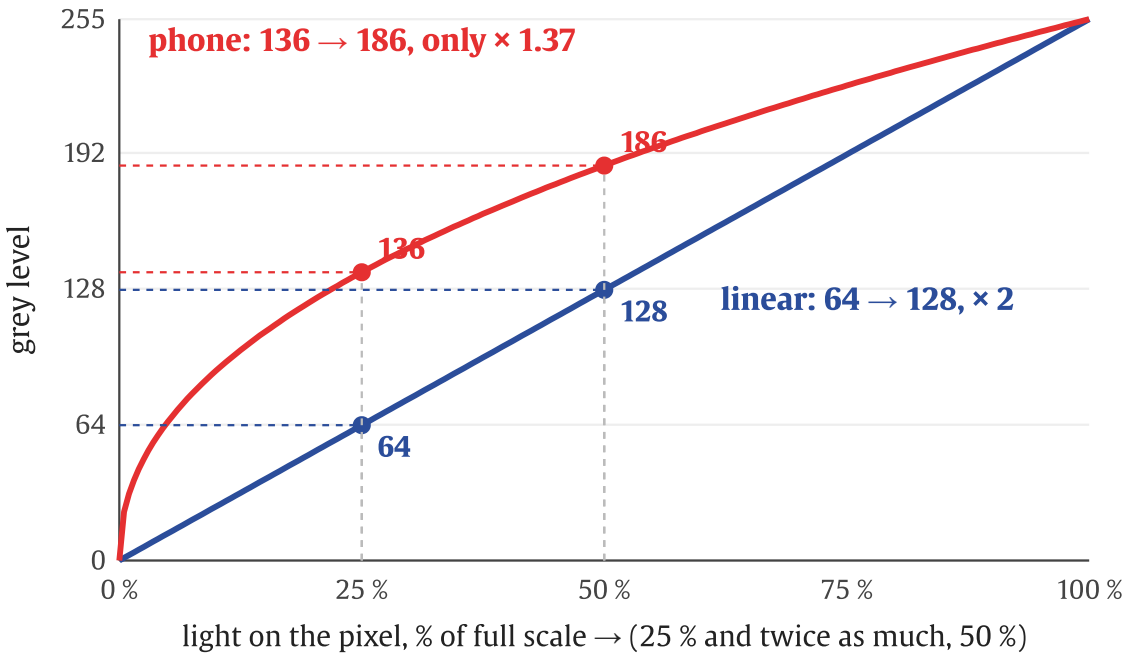
What to do

Change the lighting first, so the scene needs less range. Use HDR only for parts that stand still: the images are taken one after the other. Measure on the merged values, not on the picture adjusted for the screen.

The sensor's 2000 : 1 is a **window** on the scene's 5000 : 1, and the exposure only slides it. Here the washer comes from the short exposure, the rubber and its marking from the long one.

Linear cameras and processed images

A grey level is a measurement only if it is proportional to the light



- **What it is** — a machine-vision camera is **linear**: twice the light, twice the grey level. A phone applies a **tone curve** (gamma, about light$^{1/2.2}$) so the picture looks natural, then sharpening, noise reduction, HDR and compression.
- **Why it matters** — after the tone curve, grey levels are no longer proportional to light: a washer reflecting 8 % of full scale reads 20 on a linear camera, 81 on a phone. Brightness, contrast and noise measured on the picture are wrong; sharpening changes the edges.
- **How to measure** — a linear camera, or the phone's **raw** format; automatic exposure, white balance and HDR off, so every image is taken the same way.

In the lab

A threshold gauge is fairly robust to a tone curve — the threshold just moves; a brightness measurement is not. Part of the lab is finding out how much of a result is the phone's processing.

Recap TAKEAWAYS

Part 2 — what limits the sensor

1. **Noise:** shot, dark, read and quantisation noise are random; fixed-pattern noise is systematic. Random noise averages out, systematic noise is calibrated out.
2. **SNR ≈ $\sqrt{N}$ in good light:** four times the light for twice the SNR. Gain does not improve it.
3. **Calibration:** a dark frame and a flat field are the camera's zero and span, pixel by pixel.
4. **Saturation loses information for good:** diffuse the light, polarise, shorten the exposure; check that no peak sits at the top value.
5. **Dynamic range** = full well / read noise. Measure with linear output or raw images, never with the processed picture.

Choosing a camera for inspection

Two cases, the same questions

Question	Washers on a light box	Parts on a conveyor at 0.5 m/s
Area or line?	area-scan	area-scan, triggered by a sensor on the belt
Pixels and format	2448 × 2048, 3.45 µm, 2/3″	from field of view and smallest defect (lectures 7 and 8)
Shutter	rolling is fine: the parts are still	**global**, or a flash
Exposure	long enough to fill the well	under 82 µs, so a strong **strobe**
Colour?	monochrome	monochrome, unless colour is what is inspected
Spectrum and light	visible backlight	near-IR light and filter, to ignore the room
Bit depth	8 bits: the contrast is huge	10–12 bits if shiny and dark parts share the image
Image output	linear or raw, lossless	linear, region of interest for speed
Compare on	the **EMVA 1288** datasheet: quantum efficiency, read noise, full well, dark current, non-uniformity	

Next
Light and illumination: the part of the system that decides what the camera can see at all.

The hands-on lab
Your phone as the camera: its channels, its noise and its processing, then a gauge, then a decision on a batch.

Questions?

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github.com/LolloCappo/student-hub